Dipanwita Rano

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Education

Texas A&M University — College Station, TX
Master of Science in Computer Science ***GPA: 4.0*** — *Aug. 2024 – May 2026*
- Courses: Machine Learning, Artificial Intelligence, Computer Architecture, Deep Learning, Information Systems Retrieval, Network Security

Indian Institute of Engineering Science and Technology — Shibpur, India
Bachelor of Technology in Information Technology ***CGPA: 9.28*** — *Aug. 2020 – May 2024*
- Courses: Data Structures, Analysis of Algorithms, Database Management, Object Oriented System Design, Computer Networks, Operating Systems, Theory of Computation, Information & Systems Security, Information & Coding Theory, Software Engineering, Digital Image Processing, Compiler Design, Microprocessors

Experience

Graduate Research Assistant — May 2025 – Present
Texas A&M University — *College Station, TX*
- Cricket song analysis and synthesis, focusing on bioacoustic pattern recognition and generative modeling.
- Conduct rigorous data curation, implement multi-scale spectrogram analysis for robust cricket song classification.
- Develop conditional GAN and related generative AI models to reconstruct songs of extinct cricket species, conditioned on wing geometry and morphological features.
- Tools used: Python for data analysis and modeling, Audacity for audio processing and annotation.

Software Developer Intern — Jun. 2023 – Aug. 2023
NatWest Group — *Gurugram, India*
- Optimized the CI/CD pipeline, enhancing deployment efficiency by 15% using Agile methodologies.
- Created dynamic email reply templates triggered upon e-form submission, improving customer communication.
- Developed a dynamic automated email response system, improving response times for submitted e-forms.
- Fixed backend email queuing issue that previously blocked 20% of emails during high-load bottleneck cases.
- Tech Stack: HTML, CSS, JavaScript, FreeMarker Template, SpringBoot

Research Intern — May 2022 – Jul. 2022
Indian Institute of Engineering Science and Technology — *Shibpur, India*
- Built a TinyML-powered machine learning model for real-time heart failure prediction on embedded systems.
- Integrated and preprocessed three medical datasets, enhancing prediction accuracy by 20%.
- Developed a WebApp that streamed real-time sensor data from a Raspberry Pi for instant ML-based assessment.
- Tech Stack: Python, TensorFlowLite, Keras, Flask, Raspberry Pi, Neural Networks

Projects

Automated Medical Report Summarization: Fine-tuned transformer models (BART, Pegasus, T5) on the Cochrane dataset for both extractive and abstractive summarization. Achieved a 22% improvement in ROUGE-L scores via ensemble modeling, TF-IDF-based filtering, and paraphrase clustering. (Natural Language Processing, LLMs)

DeepFake Video Detection: Built a DeepFake detection pipeline using LAB color space-based pulse extraction and frequency domain filtering. Achieved 90% detection accuracy on FaceForensics++ by training CNNs on heart rate patterns using OpenCV and FFT. Tech Stack: OpenCV, SciPy, Keras, Dlib, Viola-Jones.

Popularity Bias Mitigation: Enhanced the PAAC recommender system by introducing contrastive temperature scaling and dynamic gradient reweighting based on item popularity. Found that a 55-45 popularity split maximizes NDCG on Yelp and Gowalla datasets. Improved model robustness across varying popularity distributions.

Technical Skills

Programming Languages: C++, Python, C, Java, JavaScript.
Web & App Development: HTML, CSS, JavaScript, ReactJS, Flask, SpringBoot, FreeMarker Template, Node.js.
Database Management: SQL, PLSQL, MongoDB, PostgreSQL.
Developer Tools: Git, GitHub, Docker, Postman, Jupyter Notebook, PowerBI, VS Code.
Systems & Tools: Linux, Shell Scripting, CUDA, ZSim.
Soft Skills: Software Development Life Cycle, Agile Methodologies, Data Science, Microsoft Word, Excel, PowerPoint.